

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Explain why binary search requires a sorted list, and state its complexity.

[5 marks]

$O(\log n)$

2

Compare compiled and interpreted execution, giving one advantage of each.

[5 marks]

3

Describe the purpose of a primary key in a relational database and why duplicates break it.

[4 marks]

4

Explain recursion using a worked example, and state the role of the base case.

[6 marks]

5

Identify the logic error in the described loop and explain the correction.

[5 marks]

6

Explain how HTTPS protects data in transit, naming the mechanism involved.

[6 marks]

7

Write pseudocode for a linear search and state its worst-case time complexity.

[5 marks]

 $O(n)$

8

Explain the difference between a stack and a queue, giving one appropriate use for each.

[5 marks]
